



DR. RICHA SINGH

ASSISTANT PROFESSOR

Department of Biotechnology
SIES College of Arts, Science & Commerce (Autonomous), Sion (W), Mumbai

CONTACT

✉ richas@sies.edu.in

📍 Mumbai, India

🌐 Orcid: 0000-0001-7184-9958

EDUCATION

- o **PhD Biotechnology**
Acinetobacter sp. mediated synthesis of metal nanoparticles & their applications in biotechnology

Savitribai Phule Pune University, India
2017
- o **MSc Biotechnology**
University of Mumbai, India
First class with distinction (73.8%)
2010
- o **BSc Hons. Biotechnology**
Panjab University, India
First class with distinction (81.3%)
2008

ADDITIONAL QUALIFICATIONS

- o **CSIR-UGC NET**
Life Sciences (AIR - 333)
2010
- o **ICMR JRF**
Research Project/Scheme of ICMR
2010

PROFESSIONAL PROFILE

Dedicated and results-driven educator with over 10 years of experience in teaching and research. Holds a Ph.D. in Biotechnology with strong interdisciplinary research background spanning Nanotechnology, Microbiology and Environmental Biotechnology. Authored 25+ peer-reviewed publications, including several in high-impact journals. Honoured with multiple national level topper awards, mentoring accolades and successful acquisition of competitive fellowships. Demonstrated excellence in leading collaborative research projects, curriculum design and student mentorship. Eager to contribute to a forward-thinking academic institution through research excellence, innovative pedagogical tools and academic leadership.

UNIVERSITY RECOGNITIONS

- University of Mumbai recognized PhD guide in Biotechnology
- University of Mumbai approved teacher for MSc (By Papers) in Biotechnology
- University of Mumbai approved teacher for MSc (By Research) in Biotechnology (since 2023)

RESEARCH ACTIVITIES

- Research focus: Bionanotechnology, Environmental Biotechnology, Microbiology
- 25+ publications in national and international peer-reviewed journals with high impact factor (*list provided at the end*)
- H-index: 19; i10 index: 19; 3000+ citations
- Guided 03 postgraduate dissertation research projects leading to publications and presentations in major conferences
 - o Synthesis, optimization and biotechnological applications of silver nanoparticles (2018)
 - o The material matters: Exploring consumer practices and safety perceptions of cookware materials (2024)
 - o Exploring the potential of banana and carrot peels for dye biosorption (2025)
- Guided 07 undergraduate projects leading to presentations in conferences
- Collaborative project with Indian Women Scientists' Association
 - o Use of plants to produce bioethanol as an alternative fuel for petrol
 - o Potential of waste mangrove apple as raw material for production of bioethanol
 - o Rapid and efficient genome editing in *Staphylococcus aureus* by using an engineered CRISPR/Cas9 system
- Participation in 10+ national and international conferences
- Attended workshops on various topics – cytogenetics, intellectual property rights, microscopy, gas chromatography, etc.

ACADEMIC ROLES

- o Assistant Professor
SIES College, Sion (W), Mumbai
2018 – Present
- o Lecturer
SIES College, Sion (W), Mumbai
2017 – 2018
- o Assistant Professor
Institute of Science, Fort, Mumbai
2016 – 2017

PROFESSIONAL SERVICES

- o Visiting Faculty
Department of Biotechnology &
Department of Biophysics,
University of Mumbai

Center of Excellence in Basic
Science
(UM-DAE-CEBS), Mumbai
- o Journal Reviewer
Frontiers in Microbiology
Frontiers in Biotechnology &
Bioengineering
- o Invited Speaker
Workshops on CSIR-UGC NET
exam,
Research Methodology &
Research
Data Analysis
- o Adjudicator
Research paper and poster
presentation competitions at
various
academic institutions

TEACHING ACTIVITIES

- Courses taught: Nanotechnology, Intellectual Property Rights, Biostatistics
Immunology, Cytogenetics, Research Methodology
- UG and PG curriculum design under National Education Policy (NEP) and
Autonomous framework
- Mentor for SWAYAM-NPTEL courses since 2018 (total 10+ topper mentees)
- Academic and career related mentoring for UG/PG students
- Convener for RUSA-sponsored workshop on Bioinformatics for UG students
- Organized seminars, guest lectures, exhibitions, field projects, industrial visits,
and alumni meets
- Organized LTMT-sponsored workshops on proteomics and epigenetics for
faculty and research scholars

ADMINISTRATIVE ACTIVITIES

- Member of various institutional committees:
IQAC Sub-Committee (Attribute IX), Admissions Committee, Time-Table
Committee,
NSS, Alumni Committee, NPTEL Committee, SIES Science Journal Editorial
Committee

ADDITIONAL CERTIFICATIONS

- UGC-HRDC RUSA sponsored Orientation course
- Refresher courses on Biosciences, Advanced Research Methodology
- NPTEL-AICTE FDPs on –
 - o Human Molecular Genetics (Gold+Elite, Top 1% certificate)
 - o Tissue Engineering (Gold+Elite, Top 1% certificate)
 - o Patent Drafting for Beginners (Silver+Elite, Top 2% certificate)
- Finding Hidden Messages in DNA (Bioinformatics I) –through Coursera
- Clinical Trial Analysis and Reporting – through Internshala

ACHIEVEMENTS & AWARDS

- Consistent Top Performing Mentor for various NPTEL courses
 - Dr. A.J. Desai / Dr. L.H. Hiranandani Award for best paper presentation at 33rd
Annual Conference of Mumbai Hematology Group
 - Roll of Honour (Academics) award for securing 3rd position in University
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PUBLICATIONS

- 1 Patil SB, **Singh R** (2026). Biosorption of procion magenta and black azabache textile dyes using banana and carrot peel waste: A comparative analysis of pure and blended systems. *Int J Phytoremediation* (accepted). IF 3.1
- 2 **Singh R** (2026). Nano-adsorbent for removal of microplastics: Cross-material insights, performance and practical applications. *Next Research* (accepted).
- 3 **Singh R**, Menon T, Yoosuf S (2026). Online professional courses' impact on teaching and personal growth. *Canadian J Learning Technol* 52(1): 1-20.
- 4 Vishwakarma OV, Patil SB, **Singh R** (2025). Practices and perceptions of food contact materials: impact of education in Indian context. *Proc Indian Nat Sci Acad* 91:1351-1364. IF 1.3
- 5 **Singh R** (2024). Antibacterial activity of bio-silver nanoparticles in combination with antibiotics. *Iran J Mater Sci Engg* 21:1. IF 0.9
- 6 Raut S, **Singh R**, Lekhak U (2024). Naturally occurring nanomaterials (NONPs): A review. *Next Sustainability* 3: 100037. IF 6.0
- 7 Nadhe SB, Wadhwani SA, **Singh R**, Chopade BA (2020). Green synthesis of AuNPs by *Acinetobacter* sp. GWRVA25: Optimization, characterization and its antioxidant activity. *Front Chem* 8: Article 474. IF 4.2
- 8 Nadhe SB, **Singh R**, Wadhwani SA, Chopade BA (2019). *Acinetobacter* sp. mediated synthesis of AgNPs, its optimization, characterization and synergistic antifungal activity against *Candida albicans*. *J Appl Microbiol* 127: 445-458. IF 3.2
- 9 **Singh R**, Vora J, Nadhe S, Wadhwani SA, et al (2018). Antibacterial activities of bacteriogenic silver nanoparticles against nosocomial *Acinetobacter baumannii*. *J Nanosci Nanotechnol* 18: 3806-3815.
- 10 Wadhwani SA, Shedbalkar UU, Nadhe S, **Singh R**, et al (2018). Decolorization of textile dyes by combination of gold nanocatalysts obtained from *Acinetobacter* sp. SW30 and NaBH₄. *Env Technol Innovations* 9: 186-197. IF 7.1
- 11 **Singh R**, Jain B (2018). Catalytic activity of chemical and biological nanosilver : A comparative study. *JETIR* 5: 243-253 (ISSN – 2349-5162).
- 12 **Singh R**, Shedbalkar UU, Nadhe SB, Wadhwani SA, et al (2017). Lignin peroxidase mediated silver nanoparticle synthesis in *Acinetobacter* sp. *AMB Expr* 7: 226. IF 3.7
- 13 Wadhwani SA, Shedbalkar UU, **Singh R**, Chopade BA (2017). Biosynthesis of gold and selenium nanoparticles by purified protein from *Acinetobacter* sp. SW30. *Enzyme Microb Technol* 111: 81-86. IF 3.7
- 14 Wadhwani SA, Gorain M, Banerjee P, Shedbalkar U, **Singh R**, et al (2017). Green synthesis of selenium nanoparticles using *Acinetobacter* sp. SW30: optimization, characterization and its anticancer activity in breast cancer cells. *Int J Nanomed* 12: 6841-6855. IF 6.5
- 15 **Singh R**, Nawale L, Arkile M, Wadhwani S, et al (2016). Phytogenic silver, gold and bimetallic nanoparticles as novel antitubercular agents. *Int J Nanomed* 11: 1889-1897. IF 6.5
- 16 **Singh R**, Nadhe S, Wadhwani S, Shedbalkar U, et al (2016). Nanoparticles for control of biofilms of *Acinetobacter* species. *Materials* 9: 383. IF 3.2
- 17 Wadhwani SA, Shedbalkar UU, **Singh R**, Chopade BA (2016). Biogenic selenium nanoparticles: current status and future prospects. *Appl Microbiol Biotechnol* 100: 2555-2566. IF 4.3
- 18 Wadhwani SA, Shedbalkar UU, **Singh R**, Vashisth P, et al (2016). Kinetics of gold nanoparticles by *Acinetobacter* sp. SW30. *Indian J Microbiol* 56:439-444. IF 1.6
- 19 **Singh R**, Nawale LU, Arkile M, et al (2015). Chemical and biological metal nanoparticles as antimycobacterial agents: A comparative study. *Int J Antimicrob Agent* 46:183-188. IF 4.6
- 20 **Singh R**, Shedbalkar UU, Wadhwani SA, Chopade BA (2015). Bacteriogenic silver nanoparticles: synthesis, mechanism and applications. *Appl Microbiol Biotechnol* 99:4579-4593. IF 4.3
- 21 Shedbalkar U, **Singh R**, Wadhwani S, Gaidhani S, et al (2014). Microbial synthesis of gold nanoparticles: Current status and future prospects. *Adv Colloid Interface Science* 209:40-48. IF 19.3
- 22 Wadhwani SA, Shedbalkar UU, **Singh R**, Karve MS, et al (2014). Novel Polyhedral gold nanoparticles: Green synthesis, optimization and characterization by environmental isolate of *Acinetobacter* sp. SW30. *World J Microbiol Biotechnol* 30:2723-2731. IF 4.2
- 23 Gaidhani S, **Singh R**, Singh D, Patel U, et al (2013). Biofilm disruption activity of silver nanoparticles synthesized by *Acinetobacter calcoaceticus* PUCM 1005. *Mater Lett* 108: 324-327. IF 2.7

- 24 **Singh R**, Wagh P, Wadhwani S, Gaidhani S, et al (2013). Synthesis, optimization and characterization of silver nanoparticles from *Acinetobacter calcoaceticus* and their enhanced antibacterial activity when combined with antibiotics. *Int J Nanomed* 8: 4277-4290. IF 6.5
- 25 **Singh R**, Pradhan V, Patwardhan M, Ghosh K (2009). APO-1/FAS gene: Structural and functional characteristics in systemic lupus erythmatosus (SLE) and other autoimmune disorders. *Indian J Human Genet* 15(3):11-15.

BOOK CHAPTERS

- 1 **Singh R** (2026). Metal and metal-oxide nanoparticles: Emerging tools in cancer therapy and bioimaging. *In: Applications of Nanomaterials for Oncology*, IGI Global Publishing, NY, pp 189-230.
- 2 **Singh R**, Menon T, Yoosuf S (2026). Role of professional courses in fostering holistic development in teaching professionals. *In: Trends in Training and Development*, Higher Education Forum, India, pp 60-64.
- 3 Menon T, Yoosuf S, **Singh R** (2025). Experiential learning: A blend of MOOCs, project-based learning and internships. *In: Teaching Learning Innovations*, Higher Education Forum, India, pp 25-30.